

Complex Network Analysis

A Multi-Layer Complex Network Analysis of Lung Cancer Survival,
Treatment Efficacy, and Progression Patterns

Syed Ali Raza

August 18, 2025

Contents

1	Introduction	1
2	Objective	1
3	Dataset	1
3.1	Preprocessing	2
4	Methodology	3
4.1	Geographical and Clinical Patterns of Survival and Treatment	3
4.2	Patient Similarity Network Construction	4
4.3	Community Detection and Survival Analysis	5
4.4	Co-occurrence Network of Comorbidities and Lifestyle Risk Factors	8
4.5	Centrality Analysis of the Co-occurrence Network	9
4.6	Patient Progression DAG Construction	11
4.7	Spectral Analysis of Patient Similarity Networks by Survival Status	11
4.8	Comparison of Treatment Effectiveness by Cancer Stage	13
4.9	Multiplex Network Construction and Analysis	14
4.10	Random Forest Classification of Survival Outcome	16
5	Results	17
6	Discussion and Conclusion	17
A	Colab Notebook Link	19

1 Introduction

Network science, often used to analyze complex systems like transportation and communication, provides a powerful framework for understanding health data. In this project, I will apply network theory to the lung cancer dataset by treating patients as nodes and model the relationships between them on the basis of clinical similarity, shared treatment and risk factors. The objective is to apply complex network theory analysis and investigate the underlying structure of patient outcomes.

The synthetic dataset of lung cancer which mimics the real-world dataset was used in the project to explore the connection between the patient profile and related survival patterns. The construction of the patient similarity network revealed the communities of different patients having different survival rates, and patient progression Directed Acyclic Graph (DAG) mapped the most common journeys. Finally, a random forest model was trained to identify the features for predicting survival, highlighting the significant role of patient comorbidities.

2 Objective

The primary objective of the projects are:

- To build a patient **similarity network** where each node represents the patients and each edge represents the how similar with respect to clinical features patients are to each other.
- To analyze the community by the Louvain method and find the community with the highest survival and the lowest survival rates. Then analyze which key factor is prominent in both communities.
- To explore how comorbidities and lifestyle risk factors appear together in patients. Which risk factor is the most common and perform a centrality analysis to check which factor is the most influential.
- Build a DAG showing the pathway from the cancer stage to patient outcome under various treatment types. This graph will help determine which treatments are most effective for any given stage of cancer.
- In order to find differences in network connectivity and clustering patterns that could indicate clinically significant differences in patient profiles and risk factors, spectral analysis will be used to examine structural differences in patient similarity networks between survivors and deceased patients.
- To analyze the two layer multiplex patient network was built to capture both clinical similarity and shared treatment relationships.
- To build the predictive machine learning model to identify the most significant clinical features associated with patient survival.

3 Dataset

Lung cancer dataset is taken from Kaggle [Saxena \(2023\)](#). to perform the analysis. It contains 890,000 patient with 18 features, which fall into the following categories:

- **Demographic details:** age, gender, and country.
- **Clinical and risk factors:** body mass index (BMI), cholesterol level, smoking status, family history, cancer stage at diagnosis, and binary flags for important comorbidities (cirrhosis, hypertension, asthma, and other cancers).
- **Intervention and results data:** the final binary survival status and the type of treatment administered.

A preliminary inspection confirmed that the dataset is complete, and has no missing values in it. It ensures a clear basis for the analysis. The data underwent a preprocessing step to properly format numerical and categorical features prior to network construction.

3.1 Preprocessing

The preprocessing was started after confirming that the dataset is complete and has no missing value to get it ready for network analyses. As I had to perform the cosine similarity and machine learning, I needed the data to be in numerical form. So to do that, I applied one-hot encoding, which converted features into a binary vector format. Doing that allowed me to calculate the cosine similarity and also to train a classifier. For the Risk-Factor Network, I took a different approach. I selected only the one-hot encoded `smoking_status` columns and the binary comorbidity columns (such as hypertension and asthma). From this subset, I calculated co-occurrence counts, which were later used as weights in the risk network.

Similarly in the Patient Progression DAG instead of encoding, I worked with aggregated data. I sorted the original DataFrame by `cancer_stage`, `treatment_type`, and `survived` status, and then counted how many patients followed each unique pathway.

4 Methodology

The methodology was informed by lecture content from Prof. Daniel Remondini [Remondini \(2025\)](#).

4.1 Geographical and Clinical Patterns of Survival and Treatment

We started our analysis by exploring how lung cancer survival rate differs by country to highlight geographical affects in patient outcome. Figure 1 shows a bar plot of survival rates across different countries. The differences in the survival rate of each country could be because of different healthcare system, patient demographics and the availability of treatment.

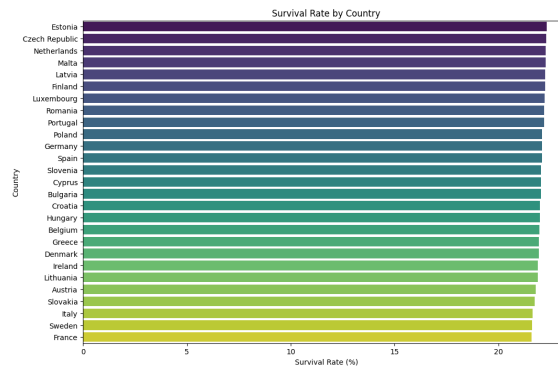


Figure 1: Survival Rate by Country (%).

As we can see in the figure 1 some countries has higher survival rate, the survival rate is higher can be because of diagnosing the cancer at early or because of better health care. It is worth doing analysis why those countries are doing better to improve the survival rate in the other countries.

In the next step, we analyze the survival rate by cancer stage and the treatment given to the patients and see how different treatments given at different stages changes the survival rate. It is shown in the Figure 2 as a heatmap of survival percentages.

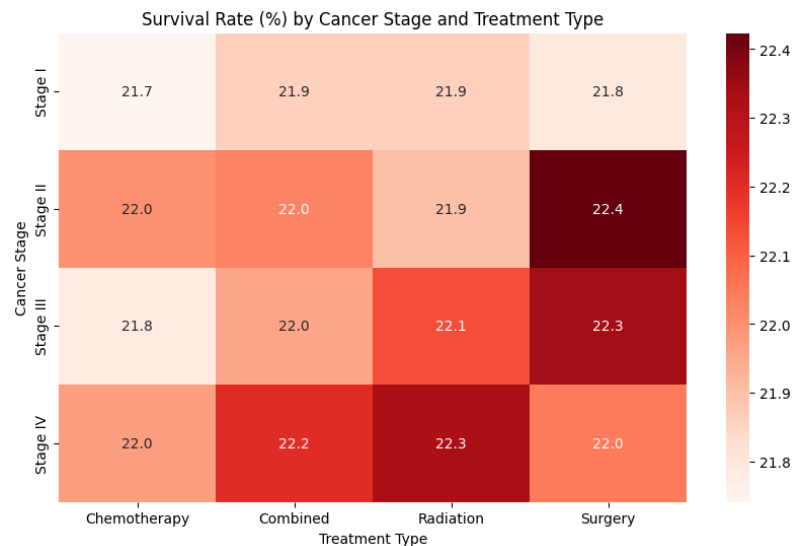


Figure 2: Survival Rate (%) by Cancer Stage and Treatment Type.

To determine whether these differences could be explained by patient characteristics, we compared

the five countries with the highest survival rates to those with the lowest survival rates, as shown in Table 1.

Countries with high survival rate	Countries with low survival rate
Estonia	Austria
Czech Republic	Slovakia
Netherlands	Italy
Malta	Sweden
Latvia	France

Table 1: Countries with highest and lowest survival rates

We performed independent t-tests for numerical variables such as age, BMI, and cancer stage, and chi-squared tests for categorical variables. None of these factors showed statistically significant differences (all $p > 0.20$), suggesting that patient-level features in this dataset do not explain the survival gap. This likely reflects the synthetic nature of the data, which does not capture real-world influences such as healthcare systems or access to treatment.

As we can see in the heatmap, usually the patients with lower cancer stage with the treatment like surgery and combined has higher survival rate then the patients with higher cancer stage. This highlight that detecting the cancer at early stage with proper treatment can lead to higher survival rates.

Finally, to provide a geographic overview combining multiple factors, I generated a choropleth map of Europe (Figure 3). The colors on this maps at different countries shows the survival rate and also combines the information about the treatment duration time and dominant treatment. In the figure here , you get only survival rate information but the colab python code , you click on the map and you get the other two informations.

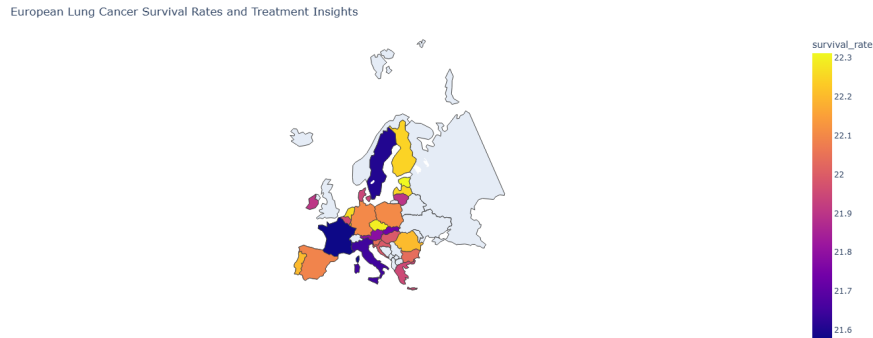


Figure 3: European Lung Cancer Survival Rates and Treatment Insights: survival rate, dominant treatment, and average treatment duration by country.

This map can help policymakers and healthcare providers to decide which country to follow to improve the treatment of lung cancer. The geography, medical decisions and patient results are interconnected and looking at them can disclose important patterns.

4.2 Patient Similarity Network Construction

In the patient similarity network, each patient is represented as a node, and edges are drawn based on how similar their feature profiles are. To prepare the data, I first created a **one-hot encoded feature matrix** that included gender, cancer stage, comorbidities, smoking status, and treatment type. Continuous variables such as age or the actual survival outcome were excluded to focus solely on shared categorical profiles. To get the similarity between patient, I used **cosine similarity**. I allowed to connect only those patients who share the similarity greater than 85 percent. I selected randomly 500 patients from the dataset to avoid the mess in the network visualization. That small dataset contains a wide variety of patients type to make easier to see clear clusters in the network.

The resulting undirected graph contains **500 nodes** and **846 edges** (Table 2). The graph has **846 edges** which means that each patient is connected to several other patients based on

the similarity and leads to communities within the network. Identifying patients in the same community can be helpful in making treatment plan for each patient.

In Figure 4, the constructed network is visualized, showing clear clusters of patients with similar clinical characteristics.

Table 2: Network Information for the Patient Similarity Graph

Metric	Value
Number of nodes	500
Number of edges	846

Patient Similarity Network (Sample of 500, Threshold=0.85)

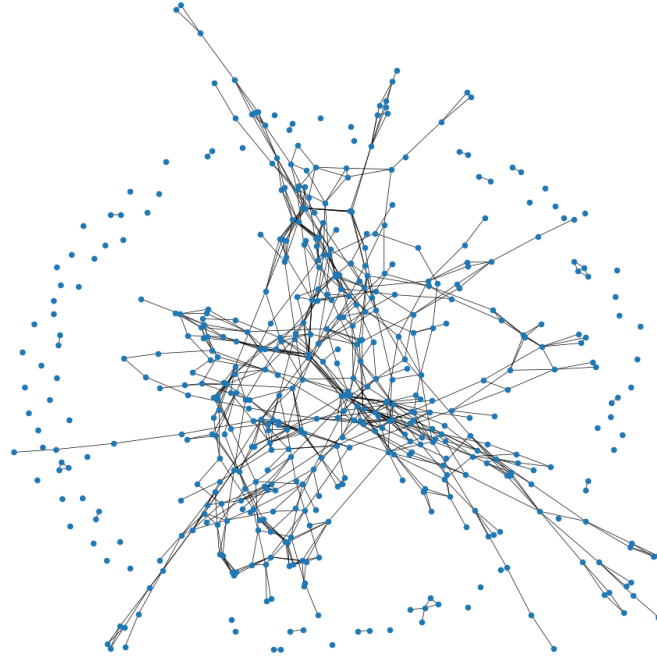


Figure 4: The Patient Similarity Network constructed from a sample of 500 patients. Each node represents a patient, and an edge indicates a cosine similarity score greater than 0.85 between their clinical profiles.

4.3 Community Detection and Survival Analysis

After building the patient similarity network, as discussed in the previous section, each patient is connected to several others, forming a cluster. In this I applied the **Louvain method** to group the patients which are very similar to each other than the others. In this way, I made the communities of patient who share the same clinical features. The communities detected by the Louvain method are shown in Figure 5.

Patient Communities Detected by Louvain Method

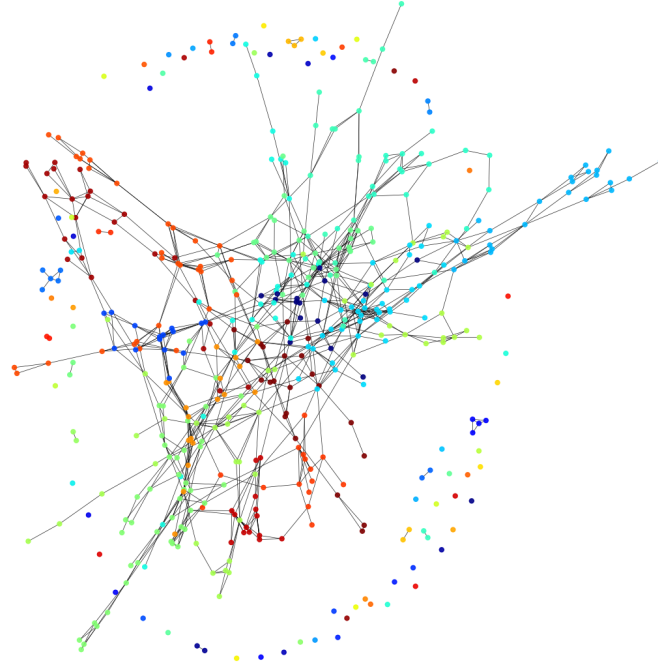


Figure 5: Patient similarity network with nodes colored by community assignment. Communities were detected using the Louvain method.

To check if the communities make sense clinically, I calculated the average survival rate of the community from the survival of patient contains in that community. This allows me to to compare the survival rate of different communities.

As shown in Figure 6 and detailed in Table 3, there was a clear variation in survival rates across communities, confirming that the detected groups are clinically meaningful.

I also plotted the barplot for communities with more than five patients. Figure 6 shows the variation in survival rates, while Table 3 presents the quantitative details.

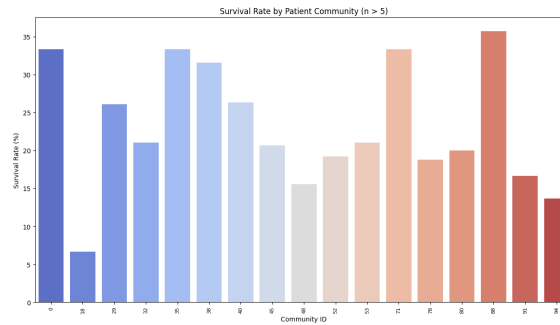


Figure 6: Comparison of survival rates across patient communities with more than five members. A significant variation in outcomes is visible, confirming the clinical relevance of the clusters.

Table 3: Survival rate by community ($n > 5$)

Community	Survival Rate (%)
88	35.71
0	33.33
35	33.33
71	33.33
38	31.58
40	26.32
29	26.09
32	21.05
53	21.05
45	20.69
80	20.00
52	19.23
78	18.75
91	16.67
48	15.56
94	13.64
18	6.67

To better understand the factors influencing these outcomes, I selected two communities for a comparative analysis:

- **Community 88** – the highest survival rate (35.71%)
- **Community 18** – the lowest survival rate (6.67%)

The detailed clinical profiles of these two groups are presented in Table 6 and Table 7. To further explore why the two communities exhibited such different survival outcomes, I compared the distributions of treatment types and smoking status between them. The summary of these comparisons is reported in Table 4 and Table 5.

The distribution of treatment modalities appeared relatively balanced in both groups. Community 88 which has a highest survival rate did show a dominance of one type of treatment which lead to high survival rate. This suggests that the outcomes are not influenced by therapies instead it may depends on patient-specific characteristics or underlying comorbidities.

The comparison of smoking status produced a more surprising finding. Community 18, which had the lowest survival rate, contained a larger proportion of patients who reported never smoking (40%) compared to Community 88 (29%). This goes against established clinical knowledge, since smoking is a major negative factor for lung cancer. The surprising result shows a limitation of the synthetic dataset. It does not fully reflect how lifestyle risks impact survival outcomes.

Table 4: Comparison of treatment type distribution (%) between Community 18 and Community 88

Treatment Type	Community 18	Community 88
Chemotherapy	26.67	21.43
Combined	26.67	28.57
Surgery	26.67	21.43
Radiation	20.00	28.57

A key distinction is that the high-survival group (Community 88) is predominantly composed of patients in Stage III, whereas the low-survival group (Community 18) shows a notably high prevalence of hypertension, which is even discussed in the article [Zeng et al. \(2025\)](#).

Table 5: Comparison of smoking status distribution (%) between Community 18 and Community 88

Smoking Status	Community 18	Community 88
Never Smoked	40.00	28.57
Passive Smoker	33.33	35.71
Current Smoker	20.00	14.29
Former Smoker	6.67	21.43

Table 6: Profile for Community 88 (Number of patients: 14)

Category	Details	Count / Value
Gender	Female / Male	9 / 5
Cancer Stage	Stage III / II / I	8 / 3 / 3
Treatment Type	Radiation / Combined / Chemotherapy / Surgery	4 / 4 / 3 / 3
Smoking Status	Passive / Never / Former / Current	5 / 4 / 3 / 2
Family History	Yes / No	6 / 8
Hypertension	Yes / No	8 / 6
Asthma	Yes / No	6 / 8
Cirrhosis	Yes / No	2 / 12
Other Cancer	Yes / No	1 / 13
Age (years)	Mean $\pm$ SD (Min–Max)	53.29 $\pm$ 10.62 (34–66)
BMI	Mean $\pm$ SD (Min–Max)	32.47 $\pm$ 8.34 (21.10–44.20)

Table 7: Profile for Community 18 (Number of patients: 15)

Category	Details	Count / Value
Gender	Female / Male	10 / 5
Cancer Stage	Stage II / I / IV / III	4 / 4 / 4 / 3
Treatment Type	Chemotherapy / Combined / Surgery / Radiation	4 / 4 / 4 / 3
Smoking Status	Never / Passive / Current / Former	6 / 5 / 3 / 1
Family History	Yes / No	5 / 10
Hypertension	Yes / No	14 / 1
Asthma	Yes / No	8 / 7
Cirrhosis	Yes / No	6 / 9
Other Cancer	Yes / No	1 / 14
Age (years)	Mean $\pm$ SD (Min–Max)	54.73 $\pm$ 8.88 (35–68)
BMI	Mean $\pm$ SD (Min–Max)	32.04 $\pm$ 6.72 (22.10–43.00)

4.4 Co-occurrence Network of Comorbidities and Lifestyle Risk Factors

We constructed the **co-occurrence network** to investigate how different health problems and lifestyle risks appear together in patients. In this network

- **Nodes:** It represents a specific risk factor (e.g., hypertension, asthma, current smoker).
- **Edge:** It connects two nodes if those two risk factors often appear together in the same patients.

In Figure 7, the thickness of nodes shows the frequency of patients sharing that problem.

As explained earlier, the thicker the edge, the stronger the co-occurrence. The Figure 7 shows the most common risk factor is the hypertension and co-occur with smoking and asthma.

To further quantify the complexity and heterogeneity of this risk-factor network, the Shannon entropy of the node degree distribution was calculated. The degree of each node corresponds to the number of other risk factors it co-occurs with among patients. The degree entropy of **0.9911 bits** shows that risk factors do not cluster around a small number of dominant factors, but rather

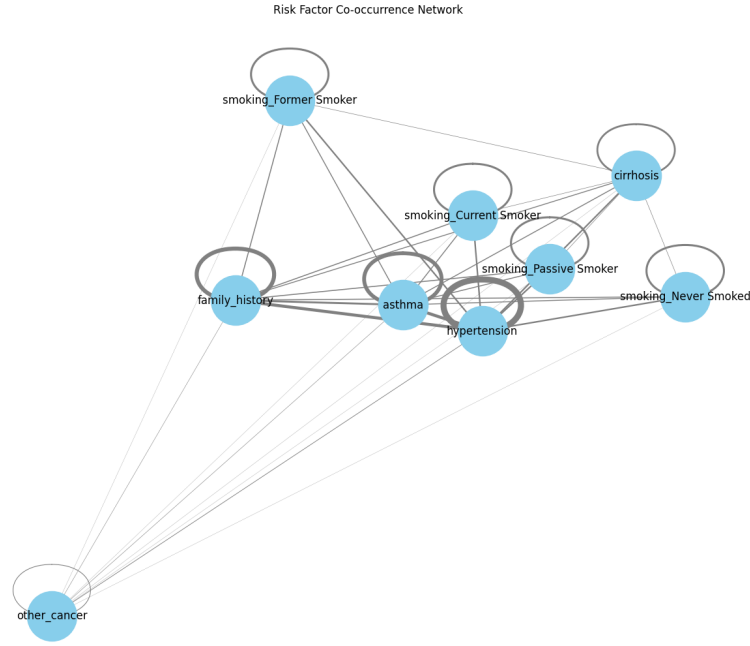


Figure 7: The Risk-Factor Co-occurrence Network. Nodes represent risk factors, and the weight of an edge corresponds to the number of patients who have both connected factors. The thickness of a node’s self-loop indicates the total prevalence of that factor.

co-occur in a wide variety of unpredictable ways. This implies that patient risk profiles are complex and consist of a variety of connected elements, proving the significance of broad medical assessments and individual therapy treatments.

4.5 Centrality Analysis of the Co-occurrence Network

The structure of the co-occurrence network can be looked at in detail by using several centrality measures. In this work, the main ones considered are **Degree**, **Betweenness**, **Closeness**, and **PageRank**.

Degree reflects how many direct links a node has, which basically shows how often a risk factor appears together with others.

Betweenness shows how often a factor sits in between and acts as a bridge for different parts of the network.

Closeness refers to how quickly a factor can reach all the other nodes in the network.

Finally, **PageRank** takes into account not only the number of connections but also the importance of the connected nodes, giving a more weighted picture of influence.

The **heat map** in Figure 8 illustrates how the risk factors fall into a central cluster and a more peripheral set. This gives a clear overview, although the exact boundary between the two groups is not always sharp.

Looking at the results in Table 8 and the heat map, *hypertension*, *family history*, *asthma*, and *cirrhosis* clearly stand out with the highest degree and betweenness values, while their closeness scores are equally high. These seem to form the network’s core. Among them, *hypertension* shows the strongest PageRank. In other words, it not only co-occurs frequently but is also connected to other important nodes.

The smoking-related categories (*Current Smoker*, *Former Smoker*, *Never Smoked*, *Passive Smoker*) have very similar PageRank values. Overall their influence is balanced, but still smaller than the central conditions, which fits with their lower degree and closeness scores.

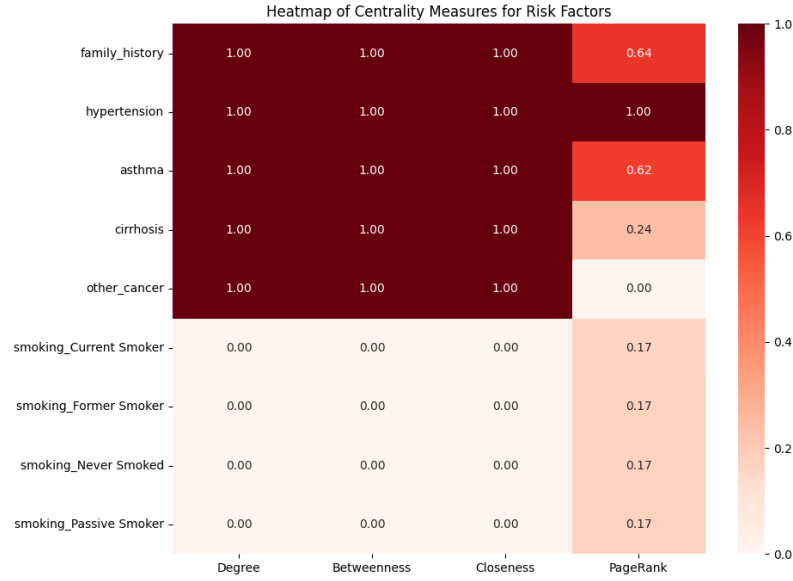


Figure 8: A heatmap representing the Risk-Factor Network’s normalized centrality scores.

Table 8: Centrality measures for comorbidities and lifestyle risk factors in the co-occurrence network.

Risk Factor	Degree	Betweenness	Closeness	PageRank
Hypertension	1.250	0.042857	1.000000	0.223328
Family History	1.250	0.042857	1.000000	0.161463
Asthma	1.250	0.042857	1.000000	0.157275
Cirrhosis	1.250	0.042857	1.000000	0.091061
Other Cancer	1.250	0.042857	1.000000	0.049369
Smoking – Passive Smoker	0.875	0.000000	0.727273	0.079545
Smoking – Never Smoked	0.875	0.000000	0.727273	0.079412
Smoking – Former Smoker	0.875	0.000000	0.727273	0.079345
Smoking – Current Smoker	0.875	0.000000	0.727273	0.079203

The centrality analysis shows more than just rankings; it also helps to understand how specific comorbidities connect with the rest of the network. Among all factors, *hypertension* is the most central, ranking highest in degree, betweenness, and PageRank, while closeness is high across the core group as a whole. This suggests that hypertension works as a “gateway” or “anchor” condition. It does not exist in isolation but ties strongly to conditions like asthma and cirrhosis. Managing hypertension properly could therefore have wider benefits, potentially improving resilience and helping patients tolerate demanding cancer treatments.

The clustering of hypertension with smoking and asthma is not unexpected, since these are well-known to co-occur in the general population. What stands out in the lung cancer group, though, is how strong these connections appear. In fact, the denser overlap suggests that these comorbidities may have a bigger role in cancer patients, perhaps through shared pathways such as chronic inflammation or immune dysfunction. Of course, confirming this would need comparisons with non-cancer groups.

Finally, the network’s degree distribution shows a high Shannon entropy of 0.9911 bits. This means patients do not follow a single dominant comorbidity profile but instead present with many different combinations. From a clinical point of view, this suggests that one-size-fits-all strategies are unlikely to work well. At the same time, the diversity of risk profiles underlines the importance of more personalized treatment plans, where each patient’s unique set of conditions is taken into account. Such tailoring is better suited to deal with the complexity revealed by the network.

4.6 Patient Progression DAG Construction

In the next part of the project, I constructed Directed Acyclic Graph (DAG) to show the pathway of patients from the time of diagnose to the final outcome. In this network the **Nodes** are:

- **Stage at diagnose**
- **Type of treatment**
- **Outcome of the patients**

The DAG is shown in Figure 9.

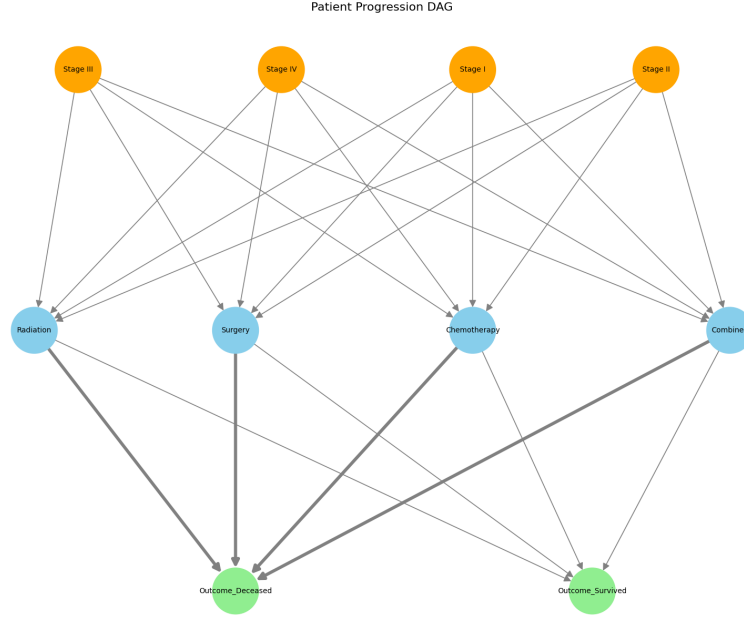


Figure 9: Directed Acyclic Graph illustrating patient progression pathways from cancer stage at diagnosis through treatment types to clinical outcomes. Edge thickness corresponds to the frequency of patients following each pathway.

The graph shows the progression from Stage $\rightarrow$ Treatment $\rightarrow$ Outcome path and the edge thickness is telling about how common was that path by different numbers of patients. Thicker edge means most common and followed by many patients. The diagram is arranged top-to-bottom (stage $\rightarrow$ treatment $\rightarrow$ outcome) so it's easy to follow the flow.

The key observation are the most of the patients ends at the outcome of deceased showing high mortality rate. This is same for almost each treatment type. There are still some treatment type which ends at the survived outcome. This visualization effectively captures both the major challenges and the critical treatment routes associated with survival.

4.7 Spectral Analysis of Patient Similarity Networks by Survival Status

The patient similarity network was constructed with lower threshold cosine similarity to investigate the structural differences between survivors and deceased patients. There are 500 nodes in the network and only those nodes are connected who has similarity greater than 0.5. We split the network into two subgroups, one survival and one deceased. Both subgroups were confirmed connected, meaning each node can be reached via some path as it was important for analysis.

The **normalized Laplacian matrices** of the survivor and deceased subgraphs were computed, and their eigenvalue spectra were obtained. The **Fiedler** value (the second smallest eigenvalue) measures the **connectivity** and **cohesivity** of the network. The fiedler value for both subgroups are shown in the Table 9.

Table 9: Fiedler Values of Survivor and Deceased Patient Similarity Networks

Patient Group	Fiedler Value
Survivors	0.5327
Deceased	0.5812

Figure 10 displays the eigenvalue distributions for both groups using histograms and kernel density estimates.

As we can see in the table above the fiedler value for the survivor group is 0.5327, which is lower then the fiedler value of deceased group. This shows that the survivor network has less connectivity overall and makes the communities compared to the subgroup of the deceased in term of their clinical features or comorbidity patterns.

These spectral differences highlight that patient similarity networks differ in topology depending on survival status. This suggests that survivors may belong to more specific, well-defined clinical profiles, while the deceased patient group is structurally more homogeneous. These structural variations indicate that analyzing the patterns in which survivors cluster might reveal particular subgroups with favorable characteristics that could direct focused treatments or actions.

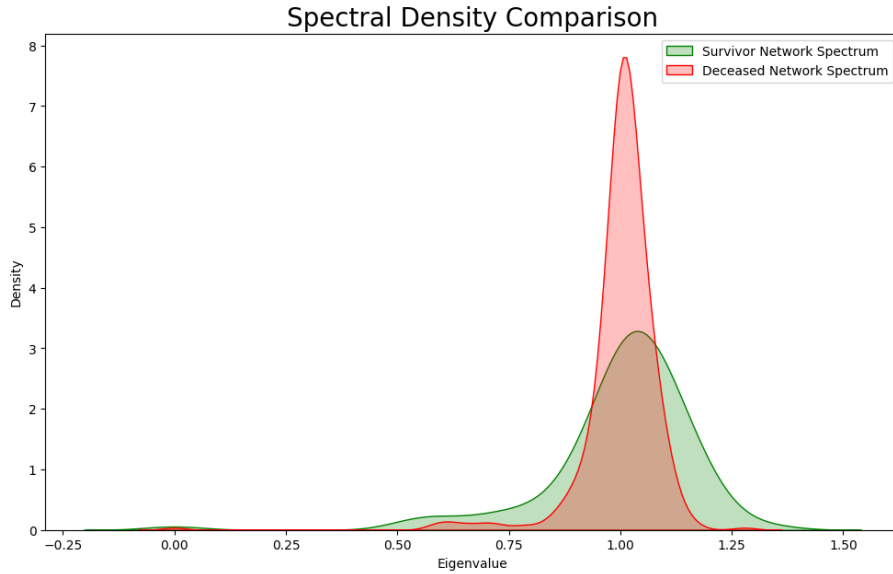
To validate the finding from the spectral analysis that survivors may form more distinct groups, network modularity was calculated for both subgraphs. The results present a detailed view of the network topology. The modularity scores are shown in the Table 10.

Network	Modularity
Survivor Network	0.2011
Deceased Network	0.1774

Table 10: Modularity scores for Survivor and Deceased networks.

The modularity score for the survivor network is higher than the deceased network. A higher modularity score indicates a stronger and more defined community structure.

The modularity results reflect the spectral analysis. This result provides strong evidence that the survivor population is more divided into separate, cohesive communities based on their clinical profiles. This supports the idea that survivors are not one uniform group, but consist of different 'favorable clinical phenotypes.' Each phenotype likely represents a unique successful clinical pathway.

**Figure 10:** Comparison of Laplacian eigenvalue spectra for survivor and deceased patient similarity networks. Histograms and kernel density estimates highlight differences in spectral density.

4.8 Comparison of Treatment Effectiveness by Cancer Stage

We looked at patient progression by analyzing survival rates for different treatment types: Chemotherapy, Combined, Radiation, and Surgery. We stratified these by cancer stage at diagnosis. Table 11 summarizes the survival rates for each treatment path. Figure 11 shows a visual comparison.

Table 11: Survival Rate (%) for Each Treatment Pathway by Cancer Stage

Cancer Stage	Chemotherapy	Combined	Radiation	Surgery
Stage I	21.74	21.86	21.86	21.79
Stage II	22.00	22.03	21.90	22.42
Stage III	21.78	21.96	22.13	22.34
Stage IV	21.97	22.20	22.33	22.05

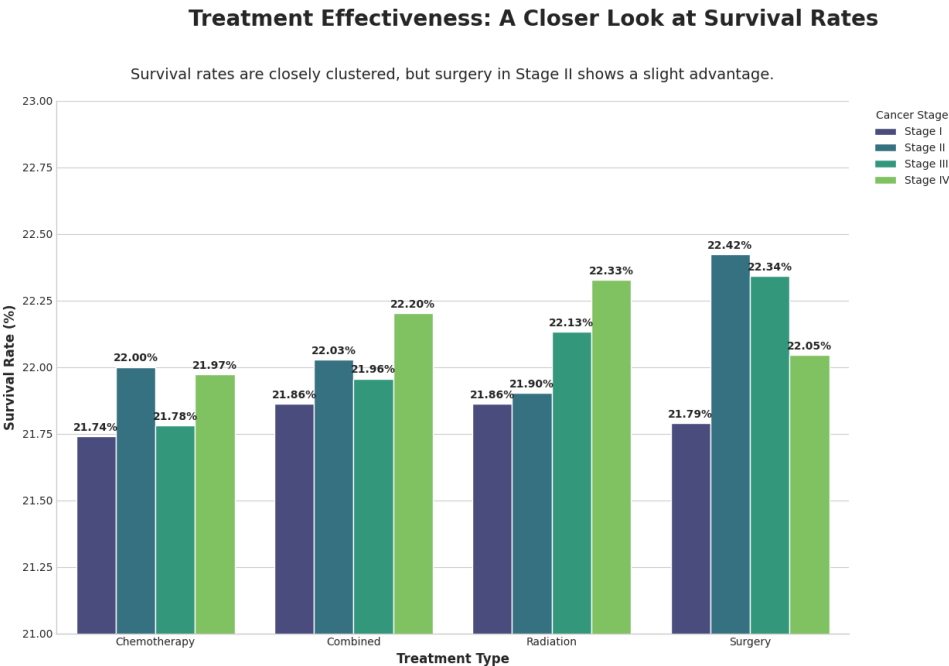


Figure 11: Comparison of survival rates across treatment types stratified by cancer stage. The bar chart highlights small but notable differences, with Surgery in Stage II showing the highest survival rate.

When we look at the results, the survival rates for the different treatments are pretty close, but a few differences do stand out. The surgery for patients in Stage II offered the best chance of survival at 22.42%. Chemotherapy and radiation appeared to be less effective for patients in both Stage I and Stage III. Overall, this suggests that the stage of the disease is the most critical factor for survival, but choosing surgery in the earlier stages could offer a slight edge.

4.9 Multiplex Network Construction and Analysis

In the next step of my project, I prepared the multiplex network to capture two types of relationships :

- (i) clinical similarity
- (ii) shared treatment.

In clinical similarity, I connected the patients who have a cosine similarity of more than 0.6 in their medical profiles. To get better visualization, I created this layer with 25 patients and have edges of 32, while the other layer was generated by linking patients receiving the same type of treatment; the number of nodes is 25 and edges are 68 in this layer.

Figure 12 shows the multiplex network where vertical dotted lines connect the same patient across both layers.

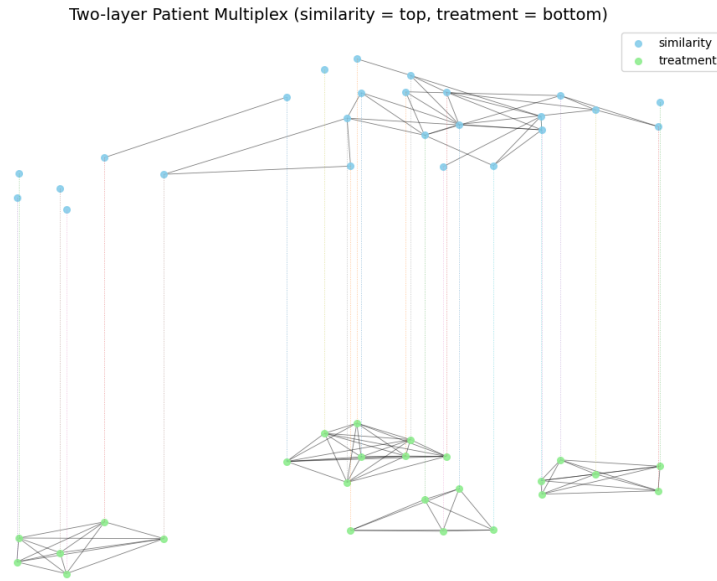


Figure 12: Visualization of the multiplex network layers where light blue node shows clinical similarity patients and green node shows shared treatment patients

In order to get more better visualization of patients who share the both clinical similarity and treatment type, I constructed the aggregated network with colored edges as shown in Figure 13. The blue color of edges is showing only clinical similarity , green color is showing treatment type and the purple shows the both.

Aggregated network (edges colored by layer membership)

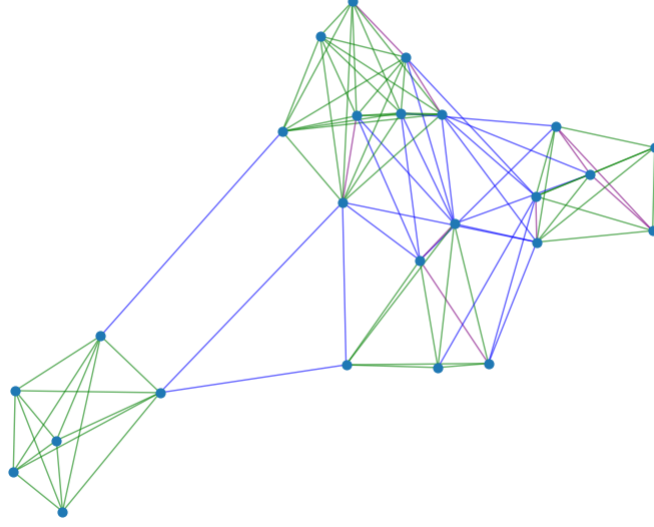


Figure 13: Visualization of the aggregated network with edge showing type of connectivity.

We can see that there is minimal overlap between two relationship manifesting clinical decision making is complex or it could be because of taking small sample of patient that's why in the next step, I will use more sample of patient to know about this more. In the table 12 below, I ran the code for 5000 patients and the overlap was still 10.48%, which manifests that making clinical decision is complex.

Table 12: Quantitative Analysis of Edge Overlap

Metric	Value
Number of Similarity Edges (Blue + Purple)	1,912,615
Number of Treatment Edges (Green + Purple)	3,126,919
Total Unique Edges in Aggregated Network	4,561,684
Number of Overlapping 'Purple' Edges	477,850
Overlap Percentage	10.48%

To see if the low overlap between clinical similarity and shared treatment layers stayed the same during different phases of the disease, the analysis was repeated on subgroups of patients. Patients were segmented into an 'Early Stage' (Stage I/II) cohort and a 'Late Stage' (Stage III/IV) cohort. The results are shown in the Table 13.

Table 13: Edge Overlap Analysis for Early-Stage and Late-Stage Patients

Metric	Early-Stage	Late-Stage
Similarity Edges	481,191	474,482
Treatment Edges	788,554	774,414
Overlapping Edges	119,947	118,419
Union Edges	1,149,798	1,130,477
Overlap Percentage	10.43%	10.48%

As shown in table above, the edge overlap percentage was 10.43% for early-stage patients and 10.48% for late-stage patients. This similar result shows that the gap between a patient's unique clinical profile and their assigned treatment plan is common in this dataset, no matter the disease stage. It supports the idea that treatment choices are influenced by a complicated mix of factors

that go beyond just clinical similarities, and these factors apply equally across all cancer stages.

4.10 Random Forest Classification of Survival Outcome

In the final stage of the project, we tested a machine learning method to see if we could predict survival outcomes directly from patient-level clinical features. We trained a Random Forest classifier on the dataset, using 70

The classifier achieved an overall accuracy of **51.04%** and an AUC-ROC score of **0.50**. These results show that the model performed like random guessing. It struggled to separate the survival and deceased classes effectively. The imbalance in the dataset strongly affected this outcome. The majority class, deceased, dominated the predictions. This is clear in the classification report, which shows the precision for deceased at 0.78 and the recall at 0.52. In contrast, the precision for survived is 0.22 and recall is 0.48.

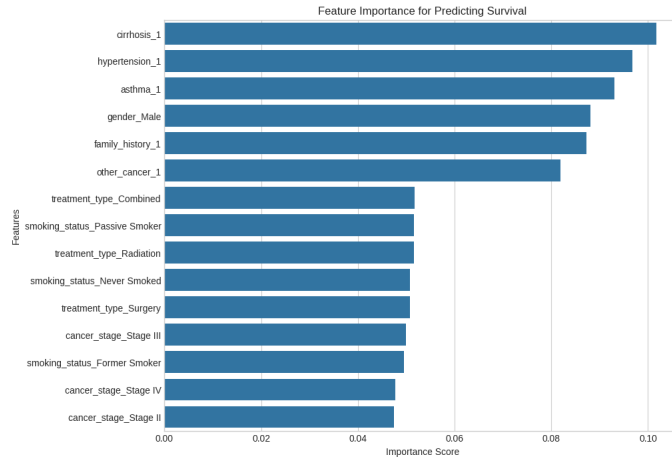


Figure 14: Feature importance scores derived from the Random Forest model. Although the classifier did not achieve predictive accuracy, the ranking of features still provides clinically relevant insights.

Although the predictive performance was weak, this result is instructive. It gives two important insights:

- **Impact of Class Imbalance:** The dataset contains a much larger number of deceased patients compared to survivors. This imbalance makes it difficult for tree-based classifiers to identify survival-specific patterns without advanced resampling techniques or cost-sensitive learning.
- **Networks vs. Machine Learning:** While the Random Forest failed to capture meaningful survival signals from independent features, the network-based analyses (community detection, centrality measures, spectral patterns, and multiplex structures) were able to uncover clinically meaningful subgroups and influential risk factors. This contrast emphasizes that survival in heterogeneous medical datasets may be better understood through *relational structures* in networks rather than standard tabular feature classifiers.

In future work, other strategies could be used to improve predictive modeling. Overall, the weak performance of the Random Forest classifier does not weaken the study. Instead, it supports the conclusion that network approaches capture higher-order interactions between patients, comorbidities, and treatments. These interactions are not easily seen through feature-based machine learning models.

5 Results

In the section, I will present the important results, which are summarized in Table 14, I got using network analysis related to lung cancer treatment outcomes, patient similarities, and risk profiles.

Table 14: Summary of Key Quantitative Results from Multi-Layer Network Analysis

Category	Description	Value / Example
Survival Rates by Community	Highest Survival (Community 88)	35.71%
	Lowest Survival (Community 18)	6.67%
Centrality Metrics (Key Risk Factors)	Hypertension (Degree)	1.25
	Family History (PageRank)	0.161
	Asthma (Closeness)	1.00
	Cirrhosis (Betweenness)	0.043
Network Entropy	Shannon Entropy of Risk-Factor Network	0.9911 bits
Spectral Analysis	Fiedler Value (Survivors)	0.5327
	Fiedler Value (Deceased)	0.5812
Number of Overlapping 'Purple' Edges Overlap Percentage		477,850
		10.48%
Predictive Modeling	Random Forest Accuracy	51.04%

As discussed in the above section related to methodology, patients were grouped into communities and showed significant differences in survival outcome. As presented in the table, community 88 has a higher survival rate than community 18. This shows that patients with similar profiles tend to show similar outcome.

In the risk-factor network analysis, hypertension appeared the most central comorbidity across all centrality measures while family history had a relatively high PageRank indicating its importance in the network. Asthma is connected to other factors, as shown by its highest closeness centrality measure. Cirrhosis has higher occurrence frequency along paths that connect various conditions evident by its highest betweenness. These measures indicate that certain comorbidities play a more influential role in patient health profiles. The higher Shannon entropy showed the complexity and diversity of the network.

As discussed earlier, higher fiedler value shows less distinct groups in the network. So we can see in the table deceased patient network has the higher fiedler value showing its uniformity compared to survivors network with respect to clinical or treatment profiles.

Only 10.48% of patient pairs share both clinical and treatment similarity, as the above table demonstrates. Given that treatment decisions do not always match with clinical feature similarity, this relatively small overlap highlights the complexity of making clinical decisions.

In the last, we trained the random forest classifier for predicting the survival status based on clinical feature. The most important feature for the model was cirrhosis.

6 Discussion and Conclusion

This project demonstrated how complex network analysis can work with a large synthetic dataset of lung cancer patients. It helps to reveal structural patterns that standard statistical methods might not easily find. By creating and examining different types of networks, the research explored survival outcomes, treatment effectiveness, and the connections between comorbidities from a broader perspective.

The patient similarity network revealed significant groups, with different survival rates among them. This confirmed that people with similar clinical profiles often group together and have similar prognoses. Hypertension and related health issues emerged as major risk factors. This shows

how comorbidities greatly impact patient outcomes. The co-occurrence and centrality analyses also showed how some lifestyle and medical risk factors strengthen each other within the larger community.

The spectral analysis of survivor and deceased networks highlighted structural differences between the two groups, with the deceased network showing higher overall connectivity. This shows that survivors group into more specific, clear profiles, while deceased patients have broader, similar risk features. Multiplex analysis, which combined similarity and treatment layers, showed that the overlap between these areas is limited. This highlights the complexity of clinical decision-making.

In contrast, the Random Forest model had difficulty reaching good predictive performance, with an accuracy of about 51% and an AUC of approximately 0.50. Instead of seeing this as a setback, it highlights an important insight. Traditional machine learning methods used on imbalanced, high-dimensional clinical data may struggle to pick up on subtle interactions. On the other hand, network-based methods showed they can reveal structural signals that feature-level classifiers fail to identify. This suggests that our way of looking at connections is a great fit for medicine, where health is often determined by a web of interconnected factors rather than single, isolated issues.

Our study shows that this network-based approach provides new ways to group patients, understand how their different conditions interact, and create better treatment plans. Although we used a simulated dataset, which means we can't apply these results in a clinic tomorrow, we have built a solid framework for real-world use.

The next step is to apply this method to actual patient records. We will include more layers of information, such as genetics or life circumstances. We also need to examine how these health networks change over time. This will help us better understand disease progression. This is how we can turn a promising theoretical model into a practical tool for personalized medicine.

A Colab Notebook Link

The full code used for data preprocessing, network analysis, and modeling is available in the following Google Colab notebook:

<https://colab.research.google.com/drive/1wBH33E0Hd31ewTT5BCEu7GBiFwqPlRXt?usp=sharing>

References

Remondini D., 2025, Lecture Notes on Complex Networks

Saxena K., 2023, Lung Cancer Dataset, <https://www.kaggle.com/datasets/khwaishsaxena/lung-cancer-dataset>

Zeng J., Shi D., He D., Dong W., Yang Z., Chen Y., 2025, *Discover Oncology*, 16, 1548